HUMPHREY VIRTUAL FARMS LLC // LEVEL-5 SOVEREIGN C2 ENCLAVE

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CLASSIFICATION: SOVEREIGN LEVEL-5 EXECUTIVE DELIVERABLE // PUBLIC/PRIVATE PARITY

Project Ebony v8.0.0: Deterministic Zero-Cloud OT/SCADA Interlock Testbed & WORM Merkl

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DOD CDAO TRADEWINDS SOLUTIONS MARKETPLACE // v8.0.0 GROUND-UP RESUBMISSION DOSSIER

NEW TITLE: Project Ebony v8.0.0: Deterministic Zero-Cloud OT/SCADA Interlock Testbed & WORM Merkle Provenance Architecture (TRL 4 Assessor-Aligned Resubmission)

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PART 1: NEW PORTAL ABSTRACT (COPY & PASTE INTO TRADEWINDS SUBMISSION FORM)

[OCT 2026 v8.0.0 RESUBMISSION DELTA: Explicitly resolves all prior 9-26-3703/10-26-4046 Assessor Feedback with new 4-Pillar Rubric proof, TRL 4 software-vs-hardware target separation, & a 5-vendor AoA matrix.] 1. PROBLEM: Per GAO-21-81/GAO-24-106137, DoD’s 500+ installations & $4.2B+ OT energy systems rely on digital PLCs scanning at 50-200ms. Per CISA AA23-144A/AA24-038A (Volt Typhoon) & TRISIS, adversaries corrupt PLC logic while spoofing HMIs. Since high-voltage arc-flash occurs in <4.16ms & thermal excursions in 6.0ms, 50-200ms scans are 8x-33x too slow. 2. SOLUTION (TRL 4 TESTBED): Project Ebony v8.0.0 runs 100% offline on 127.0.0.1 loopback (0% cloud, 636 C2 modules, 85/85 unit tests passed, 17/17 Red-Team vectors blocked, 110-control NIST 800-171/800-82 mapping). Brain 1 models <50us safety loops, 2.04us Modbus FC05 trips & a 36.75ns SiC MOSFET cutoff target; Brain 2 runs local NDE/PCI diagnostics (0 actuator wire); Brain 3 seals a 52-table SQLite SHA-256 WORM DAG. 3. STATE OF THE ART (AoA): Outclasses Siemens S7-1500/Rockwell (50-200ms, 150-450W), Triconex SIS (10-50ms), passive IDS Dragos/Nozomi (0 cutoff authority, 300-1200W), & Jev Cloud AI (>250ms RTT) via a modeled 36.75ns cutoff (1,360,544x faster than 50ms PLCs) at 28.5W on 24V DC. 4. TRANSITION: Unbundled 26-CLIN Console (CLIN 0001 Pilot $2,450/mo; 1-Node Bundle $895K/yr) & $1.485M 12-Mo Phase-1 Prototype OTA (10 USC 4022, $150K M1A) -> 4022(f) production (100% DFARS 252.227-7018 IP).

PART 2: 4-QUADRANT RESUBMISSION DELTA MATRIX (RESOLVING ALL PRIOR ASSESSOR FEEDBACK)

[DELTA-01] Prior Critique (Criterion 1): Quantify DoD OT scale, cite federal threat advisories, and prove why legacy PLC scan rates fail.

-> NEW IN v8.0.0: Anchored in GAO-21-81 & GAO-24-106137 (500+ installations, 300k+ buildings, $4.2B+ OT energy) and Joint CISA/NSA/DoD Advisories AA23-144A & AA24-038A (Volt Typhoon) plus TRISIS. Demonstrated on an animated physics timeline that <4.16ms arc-flash backfeeds and 6.0ms excursions outrun 50-200ms PLCs by 8x to 33x.

[DELTA-02] Prior Critique (Criterion 2): Separate TRL 4 software emulation from TRL 6/7 hardware claims and verify zero cloud dependency.

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-> NEW IN v8.0.0: Strictly separated our live TRL 4 local loopback software testbed (127.0.0.1, 0% cloud, 636 C2 modules, 85/85 passing unit test suites, 17/17 Red-Team blocks, 52 SQLite WORM tables, 1,420 Academy Qs) from our modeled Phase-1 OTA hardware fabrication targets (36.75ns SiC MOSFET cutoff, <50us Cortex-M7 loop, 28.5W SWaP).

[DELTA-03] Prior Critique (Criterion 3): Provide a quantitative Analysis of Alternatives (AoA) benchmarking Ebony against named competitors.
-> NEW IN v8.0.0: Added a dedicated 5-Vendor Quantitative AoA Matrix & SWaP Comparison Chart contrasting Project Ebony against Siemens S7-1500/Rockwell ControlLogix (50-200ms, 150-450W), Schneider Triconex SIS (10-50ms, 200-600W), Dragos/Nozomi passive IDS (0 physical cutoff authority, 300-1,200W), and Jev Cloud AI (>250ms API RTT).

[DELTA-04] Prior Critique (Criterion 4): Present an audit-proof commercialization roadmap, per-node pricing, and 10 U.S.C. 4022 OTA schedule.
-> NEW IN v8.0.0: Scrubbed all ambiguous deployment language; presented our Red-Team-certified client evaluation build, staged civil bridge NDE blueprints ($250/sortie drone model vs $4,600 snooper trucks), 26-Module CLIN Price Schedule (CLIN 0001-0026), and a 4-milestone $1.485M Phase-1 Prototype OTA Gantt schedule ($150K M1A Mobilization).

PART 3: 5-VENDOR QUANTITATIVE ANALYSIS OF ALTERNATIVES (COMPETITOR MATRIX)

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* METRIC: Primary Architecture
- Project Ebony v8.0.0 (1AHA8) : Tri-Brain Software Testbed (TRL 4) + Modeled 36.75ns Analog Guillotine
- Siemens / Rockwell PLCs : Digital OS/RTOS PLC (5-20Hz Cyclic Scan)
- Dragos / Nozomi Guardian : Passive Network IDS / Cloud-Hybrid SIEM
- Schneider Triconex SIS : Triple-Modular Redundant Digital PLC
- Cloud AI Wrappers (Jev) : Cloud API Classification Wrapper
* METRIC: Cutoff Latency (Design Target vs Legacy)
- Project Ebony v8.0.0 (1AHA8) : 36.75 ns (Modeled SPICE Target) / 2.04 us (Emulated Modbus FC05)
- Siemens / Rockwell PLCs : 50,000,000 - 200,000,000 ns (50 - 200 ms)
- Dragos / Nozomi Guardian : NONE (0 Physical Cutoff Authority)
- Schneider Triconex SIS : 10,000,000 - 50,000,000 ns (10 - 50 ms)
- Cloud AI Wrappers (Jev) : 250,000,000+ ns (> 250 ms Cloud API RTT)
* METRIC: Speed & Physics Advantage
- Project Ebony v8.0.0 (1AHA8) : Modeled 1,360,544x Faster than 50ms PLC (Targets <4.16ms Arc-Flash)

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- Siemens / Rockwell PLCs : 8x - 33x Too Slow for <4.16ms Arc-Flash / 6ms Excursion
- Dragos / Nozomi Guardian : Requires Human Operator (> 250 ms Reaction)
- Schneider Triconex SIS : Vulnerable to TRISIS/Triton Firmware Injection
- Cloud AI Wrappers (Jev) : Total Failure Under EMCON / SATCOM Severance

* METRIC: Target SWaP Power Draw

- Project Ebony v8.0.0 (1AHA8) : 28.5 W Modeled Target (< 30W on 24V DC Tactical Bus)
- Siemens / Rockwell PLCs : 150 W - 450 W (Rack + HMI + Active Cooling)
- Dragos / Nozomi Guardian : 300 W - 1,200 W (Appliance Server + Cloud)
- Schneider Triconex SIS : 200 W - 600 W (TMR Chassis)
- Cloud AI Wrappers (Jev) : Data Center GPU Cluster Required

* METRIC: Cloud Dependency (Verified Live)

- Project Ebony v8.0.0 (1AHA8) : 0% (100% Air-Gapped Localhost 127.0.0.1 Software Testbed)
- Siemens / Rockwell PLCs : Proprietary Firmware / Windows Engineering Workstation
- Dragos / Nozomi Guardian : High (Cloud Threat Feeds / Linux OS)
- Schneider Triconex SIS : Proprietary Firmware (Compromised in TRISIS)
- Cloud AI Wrappers (Jev) : 100% Dependent on External Commercial Cloud

* METRIC: WORM Cryptographic Ledger (Verified Live)

- Project Ebony v8.0.0 (1AHA8) : SHA-256 WORM Merkle DAG (52 Verified SQLite Tables on Disk)
- Siemens / Rockwell PLCs : Unauthenticated Modbus/CIP by Default
- Dragos / Nozomi Guardian : Mutable Syslog / SIEM Export
- Schneider Triconex SIS : Standard Event Buffer (Non-Merkle)
- Cloud AI Wrappers (Jev) : Mutable Cloud Database Logs

PART 4: 5-SCENE MULTI-ARCHITECTURE VIDEO SCRIPT
(EBONY_V8_CDAO_RUBRIC_VERIFICATION_20261028.mp4)

>>> SCENE 1 (00:00 - 00:52) // SEC VII.e MANDATORY RESUBMISSION SUMMARY & 4-QUADRANT **DELTA MATRIX**
RUBRIC: Section VII.e Mandatory Resubmission Gate & 4-Quadrant Assessor Feedback Resolution
SCRIPT: Per Announcement Section Seven E and the Appendix Assessment Rubric, welcome to the version 8.0.0 resubmission of Project Ebony by Jeffery Humphrey, Founder and CEO of Humphrey Virtual Farms LLC, CAGE 1AHA8. This briefing resolves all prior Docket 9-26-3703 feedback across four quadrants. Quadrant One quantifies defense operational technology risk per federal GAO audits and CISA Volt Typhoon advisories, proving 4.16-millisecond arc-flash faults outrun legacy PLCs. Quadrant Two separates our live TRL 4 air-gapped software testbed and zero-dependency SQLite retrieval security from

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our modeled Phase-1 hardware targets. Quadrant Three benchmarks Project Ebony head-to-head against Siemens, Rockwell, Triconex, Dragos, Nozomi, and cloud AI across six mission gates. Quadrant Four proves our commercial and federal traction, dual-use 26-module CLIN pricing, and 1.485 million dollar Phase-1 Prototype OTA.

>>> SCENE 2 (00:52 - 01:44) // APPENDIX RUBRIC CRITERION 1: PROBLEM ID, DOD IMPACT & <4.16ms PHYSICS GAP
RUBRIC: Appendix Rubric Criterion 1: Defining the Problem, DoD Mission Impact & Status Quo Risk
SCRIPT: Appendix Rubric Criterion One: Problem Identification and DoD Mission Impact. Per GAO audits 21-81 and 24-106137, the Department of Defense operates over 500 installations, 300,000 buildings, and 4.2 billion dollars in energy systems relying on 50 to 200 millisecond PLCs. Joint CISA and DoD Volt Typhoon advisories confirm state actors pre-position inside defense grids to corrupt PLC logic while spoofing operator HMIs. Under the status quo, arc-flash faults destroy equipment in 4.16 milliseconds and thermal excursions occur in 6.0 milliseconds. On our logarithmic timeline below, Project Ebony targets a 36.75-nanosecond cutoff and emulates a 2.04-microsecond Modbus trip--whereas legacy 50 to 200 millisecond PLCs react 8 to 33 times too late.

>>> SCENE 3 (01:44 - 02:38) // APPENDIX RUBRIC CRITERION 2: MISSION IMPACT, TRL 4 & ZERO-CLOUD SECURITY
RUBRIC: Appendix Rubric Criterion 2: Accelerating the Mission, TRL 4 Evidence & Zero-Dependency Security
SCRIPT: Appendix Rubric Criterion Two: Mission Acceleration, TRL 4 Maturity, and Zero-Dependency Security. On the left is our live 26-module C2 Console at TRL 4 on localhost 127.0.0.1 with zero cloud dependency, zero external retrieval APIs, and zero open WAN ports. Brain One runs deterministic safety logic and models our Phase-1 hardware target: a 50-microsecond loop, 2.04-microsecond Modbus trip, and 36.75-nanosecond silicon-carbide MOSFET cutoff, while Brain Two runs isolated read-only diagnostics with zero actuator wire. Brain Three executes all data retrieval locally inside our 52-table SQLite WORM enclave and seals a SHA-256 Merkle DAG in under 0.15 milliseconds. On the right, our live workstation subprocess harness verifies all 636 Python modules, 85 of 85 passing unit test suites, and 17 of 17 blocked Red-Team vectors with automatic tamper zeroization.

>>> SCENE 4 (02:38 - 03:31) // APPENDIX RUBRIC CRITERION 3: STATE OF THE ART & 5-VENDOR AoA MATRIX
RUBRIC: Appendix Rubric Criterion 3: Advancing the State of the Art (5-Vendor Quantitative AoA)
SCRIPT: Appendix Rubric Criterion Three: Advancing the State of the Art. Our Five-Vendor Quantitative Analysis of Alternatives benchmarks Project Ebony across six mission gates. Gate One and Gate Two evaluate architecture and cutoff latency: Siemens,

Rockwell, and Triconex require 10 to 200 milliseconds and Dragos has zero cutoff authority, whereas Ebony models a 36.75-nanosecond cutoff. Gate Three and Gate Four evaluate physics speed and SWaP: incumbents draw 150 to 1,200 watts and fail under TRISIS or EMCON, while Ebony is 1.36 million times faster than a 50-millisecond PLC at a 28.5-watt target on 24-volt DC. Gate Five and Gate Six verify zero-cloud independence and cryptographic provenance: Project Ebony passes all six mission gates with 100 percent local SQLite WORM Merkle verification.

>>> SCENE 5 (03:31 - 04:24) // APPENDIX RUBRIC CRITERION 4: VIABILITY, TRACTION,
26-CLINs & $1.485M OTA
RUBRIC: Appendix Rubric Criterion 4: Business Model Viability, Verified Traction, 26-CLINs & $1.485M OTA
SCRIPT: Appendix Rubric Criterion Four: Business Model Viability, Traction, and Transition. On the top Gantt schedule is our 12-month, 1.485 million dollar Phase-1 Prototype OTA under 10 U.S.C. 4022, starting with a 150,000-dollar mobilization tranche to advance from TRL 4 to TRL 6. In the middle is our unbundled dual-use 26-module CLIN schedule: CLIN 0001 Base Pilot at 2,450 dollars per month, CLIN 1004-A Full Node Bundle at 895,000 dollars per year, and CLIN 1004-B Base Fleet License. At the bottom is our verified federal and commercial traction: our filed U.S. Army Corps of Engineers ERDC White Paper, Defense Innovation Unit Solution Brief, microgrid pilot packets, civil bridge drone diagnostics at 250 dollars per sortie, and 110-control NIST mapping with 100 percent DFARS Background IP retained. Thank you.